

Restaurant Spending Analysis

Plotly Express tips dataset | Python | Quantitative Analytics

Objective

Examine how restaurant bill size varies by day, smoking status and meal period, and test whether party size is associated with total bill value.

Dataset and methods

The Plotly Express tips dataset contains 244 restaurant bills. Variables include total bill, tip, sex, smoker, day, time and party size. Analysis is conducted in Python using pandas, SciPy and statsmodels.

Statistical workflow

The analysis calculates descriptive statistics, applies a Welch independent-samples t-test comparing smokers and non-smokers, constructs bill categories for a chi-square analysis by day, and estimates an OLS regression of total bill on party size.

Assumption checks

Normality is assessed with Shapiro-Wilk tests, variance equality with Levene's test, and regression residuals with Shapiro-Wilk and Breusch-Pagan diagnostics. HC3 robust standard errors are used as a sensitivity check.

Business interpretation

Party size is a natural driver of bill value and can inform table allocation, staffing and capacity planning. Differences by day, smoking status or meal period can help identify demand patterns.

Reproducibility

The repository contains analysis.py, tips.csv, supporting visualisations and this report.